

Vispi Karkaria

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🎓 Education

Northwestern University

Ph.D. in Mechanical Engineering, GPA: 3.87/4.00

Focus: Machine Learning, Digital Twins, Scientific Computing

Honors: Winner of the Ph.D. Best Paper Presentation Award and the Inaugural Graduate Mentorship Award

Evanston, IL, USA
Sep. 2021 – March 2026

Northwestern University

Master's Level Graduate Certificate in Predictive Science, GPA: 4.00/4.00

Focus: Engineering Optimization, Artificial Intelligence, Uncertainty Quantification

Honors: Achieved 4.00/4.00 GPA in the Coursework

Evanston, IL, USA
Sep. 2022 – Sep. 2022

Northwestern University

Master's of Science in Mechanical Engineering, GPA: 3.870/4.00

Focus: Engineering Optimization, Deep learning, Machine Learning

Honors: Achieved 4.00/4.00 GPA in the Coursework

Evanston, IL, USA
Sep. 2021 – Dec. 2025

College of Engineering, Pune (COEP)

Bachelor of Engineering in Mechanical Engineering, GPA: 9.84/10.00

Graduated with Gold Medal; Ranked 1st in university

Pune, India
Jul. 2018 – Jul. 2021

Cusrow Wadia Institute of Technology

Diploma in Mechanical Engineering, Grade: 96.5%

Graduated with distinction; Received 4 Gold Medals and 11 Academic Awards

Pune, India
Aug. 2015 – May 2018

🏢 Industry Experience

Northwestern University

Graduate Research Assistant

- Developed transformer neural net and Uncertainty-Aware Reinforcement Learning frameworks, enhancing tire monitoring, extending longevity by 25%, and improving decision accuracy by 50%. This work was conducted on applied research projects in collaboration with Michelin Tires.
- Created U-LoRA for efficient machine learning updates and VR-SMOTE for data balancing, boosting F1-score by 50%.

Evanston, USA
September 2021 – Present

Epsilon Data Management LLC

Ph.D. Machine Learning Scientist Intern

- Led development of a transformer-based model that improved win-rate prediction accuracy in AdTech systems by 25%.
- Built an interpretable model with integrated uncertainty quantification (UQ), increasing bidding decision reliability by 20% and reducing failed bids by 15%.

Chicago, USA
June 2025 – September 2025

Ansys INC

Machine Learning Research Intern

- Developed a hybrid modeling framework using Transformers for time series analysis and temporal-spatial sensitivity analysis, improving predictive accuracy by 30%, with a patent applied for the machine learning algorithm developed.
- Integrated the framework into Ansys Twin AI software, reducing computational time by 25% and enhancing the scalability of the model, with the underlying technology Patented.

Evanston, USA
June 2024 – September 2024

Tata Research Development and Design Center

Machine Learning Intern

- Leveraged machine learning (Gaussian processes) to achieve a 50% reduction in material selection time for airplane wing manufacturing.
- Spearheaded a data-driven three-phase project encompassing digital twin creation, machine learning model development, and data uncertainty analysis.

Pune, India
August 2020 – August 2021

- Enhanced a village's education outcomes by 20% over two years through data-driven policy modeling using System Dynamics.
- Employed analytical methods, including the Dynamic Triangle Method, to develop data-centric Smart Village interventions.

Research Experience

Multi-Agent IIm System for Automated Optimization Problem Solving

Algorithms: GPT, Agentic AI, RAG-based Systems, Language Models

2023–Present

- Initiated the creation of an innovative API using GPT, BART, and a Retrieval-Augmented Generation (RAG) system, allowing users to input optimization problems for intelligent solutions.
- Implemented a two-stage approach where the system first selects the appropriate optimization method using RAG and then invokes it, providing efficient and accurate solutions to users.

Federated Learning Foundational Models with Attention-based Neural Operator

Algorithms: Federated Learning, Neural Operator, Attention, Foundational Model

2024–Present

- Developed an Attention-based Neural Operator, enhancing the accuracy of neural operators by 25%, and facilitating the analysis of temporal-spatial relationships.
- Adapted the Attention-based Neural Operator into a foundational model, leveraging Federated Learning for transferability and scalability across different applications with Cyber-Security in mind.

Reinforcement Learning-Transformer based Tire Resource Allocation Algorithm

Algorithms: Transformers, Sequential Learning, Reinforcement Learning

2023–Present

- Boosted real-time tire lifespan prediction accuracy by 50% using an advanced Interpretable Temporal Fusion Transformer.
- Developed a reinforcement learning framework for tire resource allocation, extending tire health by 95% using the transformer model.

Transformer-based Digital Twin framework for Time Series Optimization

Algorithms: Deep Learning, Big Data, Transformers, Bayesian Optimization

2022–2024

- Reduced prediction time in Additive Manufacturing using a transformer-based surrogate model.
- Developed a transformer-based machine learning model to quantify uncertainty, enhancing the reliability of predictions.
- Created a digital twin using a novel method named Bayesian Optimization of Time Series (BOTS) which aid in time series optimization.

Latent-Variable Constrained Bayesian Optimization (with Hyundai Motors)

Algorithms: Mixed Variable Optimization, Gaussian Processes, PyTorch

2022–2023

- Delivered a 20% rise in Hyundai Motor Company's operational efficiency and cut costs by 15% by addressing optimization challenges with LV-CBO.
- Pioneered LV-CBO to seamlessly handle both continuous and categorical variables.

Publications & All Papers on Google Scholar

- Karkaria, V., Lee, D., Chen, Y., Yu, Y., and Chen, W., 2025. "An Attention-based Spatio-Temporal Neural Operator for Physics Prediction and Discovery." *Machine Learning: Science and Technology*
- Karkaria, V., Tsai, Y-K., Chen, Y-P., and Chen, W., 2025. "An Optimization-Centric Review on Integrating Artificial Intelligence and Digital Twin Technologies in Manufacturing." *Engineering Optimization*, 57(1), pp.161–207.
- Chen, Y-P., Karkaria, V., Tsai, Y-K., Rolark, F., Quispe, D., Gao, R. X., Cao, J., and Chen, W., 2025. "Real-Time Decision-Making for Digital Twin in Additive Manufacturing with Model Predictive Control Using Time-Series Deep Neural Networks." *Journal of Manufacturing Systems*, 80, pp.412–424.
- Karkaria, V., Chen, J., Luey, C., Siuta, C., Lim, D., Radulescu, R., and Chen, W., 2024. "A Digital Twin Framework Utilizing Machine Learning for Robust Predictive Maintenance: Enhancing Tire Health Monitoring." *Journal of Computing and Information Science in Engineering*, 25, pp.01–11.
- Dewberry, N. K., AlHmoud, I., Benton, K., Suarez, D., Chen, Y-P., Karkaria, V., Tsai, Y-K., et al., 2025. "A Real-Time VR-Enabled Digital Twin Framework for Multi-User Interaction in Industry 4.0." *Manufacturing Letters*, 44, pp.1486–1497.

6. **Karkaria, V.**, Goeckner, A., Zha, R., Chen, J., Zhang, J., Zhu, Q., Cao, J., Gao, R.X. and Chen, W., 2024. "Towards a Digital Twin Framework in Additive Manufacturing: Machine Learning and Bayesian Optimization for Time Series Process Optimization." *Journal of Manufacturing Systems*, 75, pp.322-332.
7. **Karkaria, V.**, Chen, J., Siuta, C., Lim, D., Radelescu, R., & Chen, W., 2023. "A Machine Learning Based Tire Life Prediction Framework for Increasing Life of Commercial Vehicle Tires." *Journal of Mechanical Design*, 1–21.
8. van Beek, A., **Karkaria, V. N.**, and Chen, W., 2023. "Digital Twins for the Designs of Systems: A Perspective." *Structural and Multidisciplinary Optimization*, 66(3), p.49.
9. Jha, D., **Karkaria, V. N.**, Karandikar, P. B., Desai, R. S., 2022. "Statistical Modeling of Hybrid Supercapacitor." *Journal of Energy Storage*, 46, 103869. .
10. **Karkaria, V. N.**, Pradeep, K., Rohith Kumar, T., Karandikar, P. B., Mistree, F., Allen, J. K., 2021. "Statistical Analysis Of Electrode-Electrolyte-Separator To Improve Effectiveness Of Supercapacitor." *Turkish Journal of Computer and Mathematics Education*, 2(14), 1600.
11. Godse, L. S., **Karkaria, V. N.**, Bhalerao, M. J., Karandikar, P. B., and Kulkarni, N. R., 2020. "Process Based Modeling of Nylon Separator Supercapacitor." *Energy Storage*, e204. .
12. Kokate, V. L., **Karkaria, V. N.**, Karandikar, P. B., and Holmukhe, R. M., 2020. "Implementation of Design of Experiments for E-Rickshaw Range Prediction." *Journal of Advanced Research in Dynamical and Control Systems*, 12, 1219–1227. .

Conference Papers

1. **Karkaria, V. N.**, Chen, Y-P., Tsai, Y-K., Erdogan, G., Radulescu, R., & Chen, W., 2025. "AI-Driven Digital Twin Framework for Predictive Maintenance: Dynamic Decision Making and Adaptive Model Updates." In *Proceedings of the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)*, Anaheim, California, August 17–20, 2025
2. Tsai, Y-K., **Karkaria, V. N.**, Chen, Y-P., & Chen, W., 2025. "Multi-Generation Control Co-Design for Digital Twin-Enabled Systems With Deep Reinforcement Learning." In *Proceedings of the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)*, Anaheim, California, August 17–20, 2025.
3. Chen, Y-P., Tsai, Y-K., **Karkaria, V. N.**, & Chen, W., 2025. "Real-Time Decision-Making Under Uncertainty in Digital Twins: Robust Model Predictive Control Using Time-Series Deep Quantile Learning." In *Proceedings of the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)*, Anaheim, California, August 17–20, 2025.
4. **Karkaria, V.**, Das, A. K., Yadav, A., Sharma, A., Allen, J. K., & Mistree, F., 2021. "A Computational Framework for Social Entrepreneurs to Determine Policies for Sustainable Development." In *Proceedings of the ASME 2021 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)*, Vol. 3B: 47th Design Automation Conference (DAC), Virtual, August 17–19, 2021. Paper No. V03BT03A019. .
5. Shinde, S., Momin, T., **Karkaria, V. N.**, Karandikar, P. B., 2021. "Investigation of Substitute Jar Materials for Laboratory-Grade Ball Milling Machine to Process Electrode Materials for Energy Storage Devices." *IOP Conference Series: Materials Science and Engineering*, Vol. 1206, 012018. .
6. Godse, L. S., **Karkaria, V. N.**, Bhalerao, M. J., Khatua, S., Karandikar, P. B., 2019. "Electrode–Electrolyte Compatibility for Superior Performance of Supercapacitors." In *Proceedings of the International Conference on Power Electronics Applications and Technology in Present Energy Scenario (PETPES)*, IEEE, pp. 1–5. .
7. Kokate, V. L., Bankar, D. S., Holmukhe, R. M., **Karkaria, V. N.**, Karandikar, P. B., 2018. "E-Rickshaw: Present, Past and Future with Reference to Current Transportation in India." In *Proceedings of the International Conference on Recent Innovations in Electrical, Electronics & Communication Engineering (ICRIEECE)*, Vol. 1(1), pp. 1–15.
8. **Karkaria, V. N.**, Sarathchandran, V., Korgaonkar, P. R., Reddy, G. S. R., Mehta, S., Karandikar, P. B., 2018. "Review of Technologies for Making Electric Vehicles the Main Mode of Transport." In *Proceedings of*

the International Conference on Advances in Communication and Computing Technology (ICACCT), IEEE, pp. 142–148. .

9. Godse, L. S., **Karkaria, V.**, Karandikar, P. B., Kulkarni, N. R., 2017. "Innovative Methods of Ball Milling to Grind Activated Carbon as an Electrode Material for Enhancing the Performance of Ultracapacitors." In *Proceedings of the International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS)*, IEEE, pp. 2034–2041. .

Book Chapters

1. **Karkaria, V.**, Chen, J., Siuta, C., Lim, D., Radelescu, R., and Chen, W., 2024. "Tire Life Assessment for Increasing Re-Manufacturing of Commercial Vehicle Tires." In *Technology Innovation for the Circular Economy: Recycling, Remanufacturing, Design, Systems Analysis and Logistics*, pp. 599–612.

Patent Applications

1. **Karkaria, V. N.**, Bhalerao, M. J., Wankhede, P., Kokate, V. L., and Karandikar, P. B., 2019. "Economic Test Setup for EV Motor Testing Using Innovative Coupling." Indian Patent No. 201921020439.
2. **Karkaria, V. N.**, Wardhekar, S. V., Prabhune, A. A., and Shinde, H. P., 2019. "Modified Hot Water Supply System for Water and Energy Saving." Indian Patent No. 201921021898.
3. **Karkaria, V. N.**, Bhalerao, M. J., Korgaonkar, P. R., Solanke, S. R., Kamble, U., and Karandikar, P. B., 2018. "Forward-Reverse Motion Multi-Cylinder Ball Milling Machine." Indian Patent No. 201821008148.
4. **Karkaria, V. N.**, Godse, L. S., Kulkarni, N. R., Choudhary, S., Singh, V., and Karandikar, P. B., 2017. "Motor/Bike Wheel-Driven Disc-Type Ball Mill Machine for Grinding Electrode Material of Electrochemical Storage Devices." Indian Patent No. 201721004818.
5. Godse, L. S., Karandikar, P. B., **Karkaria, V. N.**, Kulkarni, N. R., Kumar, P., Sharma, P., and Gadwal, P., 2016. "Advanced Ball Mill Machine for Grinding of Electrode Material for Electrochemical Devices." Indian Patent No. 201621023599.

Selected Honors and Awards

- **President, AI@NU** — Led AI research initiatives and student engagement at Northwestern University (2025).
- **Martin Outstanding Graduate Student Award** — Recognized for academic excellence and leadership in graduate research (2025).
- **NAMRC Best Paper Award** — Awarded for a paper published in the Journal of Manufacturing Systems (JMS 2025).
- **NAMRC Student Research Award** — Recognized for a paper on digital twin frameworks in additive manufacturing (2024).
- **Inaugural Graduate Mentorship Award** — Honored for excellence in mentorship in the Mechanical Engineering Department at Northwestern University (2024).
- **NSF Fellowship Award** — Received at the 2nd International Association for Computational Mechanics (IACM) Conference (2023).
- **Predictive Science and Engineering Design Fellowship**, Northwestern University (2022–2023).
- **Walter P. Murphy Fellowship**, Northwestern University (2021–2022).
- **Gold Medalist**, University of Pune — Graduated first in undergraduate cohort (2021).
- **First Prize**, Paper Presentation Competition — Walchand College, Sangli (2021).
- **First Prize**, Project Presentation — Indian Institute of Technology (IIT) Roorkee (2021).
- **Best Research Paper Award** — International Conference on Communication, Computing, and Electronics Systems (2020).
- **First Prize**, Garrett Motion Project Competition — Army Institute of Technology (2020).
- **Second Prize**, Indian Institute of Metals Conference — College of Engineering Pune (2020).
- **First Rank**, Mechanical Engineering (Second Year) — College of Engineering Pune (2019).
- **First Prize**, Paper Presentation — Army Institute of Technology, Pune (2019).

- **Top Rank Holder**, Cusrow Wadia Institute of Technology — Awarded **4 Gold Medals** and **11 Academic Prizes** (2015–2018).
- **First Prize**, Paper Presentation — Anant Rao Pawar College of Engineering and Government Polytechnic, Pune (2018).
- **First Prize**, Diploma (First and Second Year) — Cusrow Wadia Institute of Technology (2018).

Training and Certification

- Earned certificate in Solid Works and Catia from Cusrow Wadia Institute of Technology (2018).
- Awarded certificate in Ansys and Dassault Abaqus from College of Engineering Pune (2019-2020).
- Received certificate in Systems Engineering for Industries Course from Siemens Company (2019).
- Earned certificate in Python coding from Coursera (2019).
- Earned certificate in Robotics from College of Engineering, Pune (2018).
- Received certificate in C, C++ from Disha Computer Classes (2018).
- Received certificate in Auto CAD from Cusrow Wadia Institute of Technology (2017).

Mentorship Experience

2025–Present: Ami Takamatsu — Undergraduate researcher exploring the use of genetic algorithms for engineering design optimization.

2024–2025: Christopher Leuy — Graduate mentee on a digital twin framework for remanufacturing; co-authored journal article published in *Journal of Computing and Information Science in Engineering (JCISE)*.

2022–2024: Vinamra Saxena — Master's student supervised on research in machine learning and mechanical engineering.

2022–2023: Dr. Dong Hyun Ha (Hyundai Motors) — Collaborated on applying latent-variable Bayesian optimization with trust-region methods for automotive design.

2023–2024: Arman Jain — Mentored in additive manufacturing and engineering design, with emphasis on digital twin integration.

Teaching Experience

Northwestern University

Grader, ME 341/441 – Engineering Optimization for Product Design and Manufacturing 2023–2025

- Assisted in developing quizzes and homework assignments focused on optimization algorithms and applications.
- Held regular office hours and provided project mentoring to 3–4 student teams, supporting technical development and implementation.
- Helped students debug code and interpret optimization results for course projects involving real-world design problems.

Northwestern University

Peer Mentor, CS 497 – Sports Technology 2023–2024

- Peer-mentored student groups working on machine learning applications in sports analytics.
- Advised on data preprocessing, model selection, and validation strategies for semester-long projects.

Grant Writing and Research Support

2024–2025: Assisted **Professor Wei Chen** in preparing major federal research proposals, including:

- **2025 Vannevar Bush Faculty Fellowship (VBFF)** — Supported literature review, technical framing, and alignment for project titled Foundation Models and Decisions for Digital Twins as a Learning System
- **DARPA Disruption Opportunity** — Contributed to proposal sections on foundational model development, uncertainty quantification, and digital twin integration in manufacturing